SiamOHOT: A Lightweight Dual Siamese Network for Onboard Hyperspectral Object Tracking via Joint Spatial–Spectral Knowledge Distillation

Chen Sun, Xinyu Wang, *Member, IEEE*, Zhenqi Liu, Yuting Wan, *Member, IEEE*, Liangpei Zhang, *Fellow, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— Hyperspectral object tracking is aimed at tracking targets by using both spatial information and abundant spectral information, overcoming the drawbacks of traditional RGB tracking in complex scenarios, such as low resolution or background clutter. However, the current hyperspectral object tracking methods usually have a high computational complexity, due to the huge data volume, making them difficult to apply to real-time applications on edge devices (e.g., robots, unmanned aerial vehicles, and satellites) with limited computational resources. In this article, a lightweight dual Siamese network for onboard hyperspectral object tracking—termed SiamOHOT—is proposed for real-time and onboard tracking. Specifically, a joint spatial–spectral knowledge distillation method is proposed to teach a lightweight dual Siamese tracker to learn from a deep tracker—SiamHYPER—so that the number of parameters can be compressed to improve the computational efficiency. In addition, a deep learning inference optimizer is introduced to fuse the layers with similar functions and quantify the parameters of the network, to further promote the processing speed when deployed on an embedded platform. The proposed lightweight model was verified using the 2021 WHISPERS Hyperspectral Object Tracking Challenge dataset, and achieved a superior efficiency and accuracy. In addition, a prototype system was built integrating a snapshot hyperspectral imager, the SiamOHOT tracking algorithm, and an artificial intelligence edge device (NVIDIA Jetson Xavier NX), to realize real-time imaging and tracking. The inference speed of the optimized SiamOHOT network is nearly doubled when compared to the teacher model on the prototype system.**

***Index Terms*— Dual Siamese network, hyperspectral object tracking, onboard processing, spatial–spectral distillation.**

I. Introduction

HYPERSPECTRAL object tracking is aimed at locating the position of the target of interest in each frame of a hyperspectral video, based on its spatial information and abundant spectral features [1]. Compared with RGB videos, hyperspectral videos have a higher spectral resolution, which makes it possible to acquire complete and unique spectral fingerprints of the target of interest. The introduction of spectral features can enhance the discriminative ability for distinguishing tiny or camouflaged objects, and the robustness of challenging dynamic scenes, e.g., scenes with background clutter or a low resolution. Overall, hyperspectral object tracking has great potential in various fields, such as military reconnaissance and security surveillance, and is one of the current research hotspots of hyperspectral image processing.

Hyperspectral object tracking has benefited from the development of snapshot hyperspectral imaging technology, which can obtain sequential hyperspectral data cubes at a video frame rate [2], [3], [4], [5]. However, as a new frontier technology, hyperspectral object tracking also suffers from the challenges of both the huge data rates and the "data hungry" problem [6], [7]. Specifically, the data volume to be processed per second is a function of the spatial dimension, spectral dimension, and acquisition frame rate, and is thus very large and requires the hyperspectral object tracking model to have a high efficiency. In addition, compared to RGB-based tracking, hyperspectral object tracking is faced with the "data hungry" problem, i.e., inadequate training samples, and hyperspectral video data make it very difficult to train a robust hyperspectral tracker.

Hyperspectral object tracking was first proposed by Banerjee et al. [1], utilizing the spectral angle mapper to locate the most similar regions and distinguish foreground from background. Since then, several hyperspectral object tracking methods have been developed, including mean shift-based methods [8], correlation filter-based methods [9], [10], [11], and deep learning-based methods [12], [13], [14], [15], [16], [17]. Among these different methods, the deep learning based dual Siamese network [15], [16], [17] is an advanced solution for hyperspectral object tracking with a high tracking accuracy. The dual Siamese network merges an RGB Siamese network and a hyperspectral Siamese network, where the RGB branch is pretrained using huge amounts of RGB data so as to extract spatial features from the pseudo-color data generated by the hyperspectral data. Meanwhile, the hyperspectral branch is used to extract hyperspectral semantic information for enhancing the discriminative properties of

Manuscript received 9 March 2023; revised 4 July 2023; accepted 23 July 2023. Date of publication 21 August 2023; date of current version 8 September 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903404, in part by the National Natural Science Foundation of China under Grant 42101327 and Grant 42325105, and in part by the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS) Special Research Funding. *(Corresponding author: Xinyu Wang.)*

Chen Sun, Zhenqi Liu, Yuting Wan, Liangpei Zhang, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan City, Hubei 430072, China (e-mail: sunchen@whu.edu.cn; liuzhenqi@whu.edu.cn; wanyuting@whu.edu.cn; zlp62@whu.edu.cn; zhongyanfei@whu.edu.cn).

Xinyu Wang is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, Hubei 430072, China (e-mail: wangxinyu@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3307052

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

the hyperspectral object tracking model. Generally speaking, an advanced performance is achieved by the dual Siamese tracking network, but its computational cost is still relatively high, since the number of parameters is almost twice that of the classical Siamese network with only one branch, which makes it difficult to directly apply it to some real-time processing tasks.

When hyperspectral object tracking meets embedded and edge systems, e.g., industrial robots, cars, unmanned aerial vehicles, and satellite platforms, the limited computing power puts forward higher requirements for the efficiency of the hyperspectral object tracking method, compared to some RGB trackers on edge platforms [18], [20]. Generally speaking, the mismatch between the high-resource-requirement software and the limited-resource hardware results in the following problems: 1) the large amounts of hyperspectral data incoming in real time occupy storage resources and increase the memory burden and 2) the huge number of parameters in the hyperspectral object tracking methods bring a high computational cost, which can be beyond the computing capacity of edge devices. These conflicts make it difficult for hyperspectral models to deduce in real time on edge platforms. Taking SiamHYPER [15] as an example, when it is deployed on an edge device (NVIDIA Jetson Xavier NX), we found that its inference speed decreases from 25 to 1.7 frames/s. Because the data volume of a hyperspectral image is dozens of times larger than that of an RGB image, acquiring hyperspectral images at video frame rates takes up more storage resources. In addition, to explore the rich spatial information in depth, deep hyperspectral object tracking models usually adopt a deeper backbone network, e.g., ResNet50, which takes up more storage, computational resources, and time, and as a result limits the real-time application of these models.

In this article, to address the above issues, we propose a lightweight model, where knowledge distillation is utilized to compress the backbone network and obtain the equivalent lightweight network—SiamOHOT (student)—of the SiamHYPER [15] network (teacher). The model is then deployed on the embedded platform to realize the pipeline of online processing. The knowledge distillation method obtains a similar tracking performance to the teacher network by introducing a distillation loss function so that the output of the student network approximates that of the teacher network. The spatial–spectral joint knowledge distillation (JSSKD) method designed in this study obtains the distillation loss by weighting the mean squared error (mse) loss of the hyperspectral branch classification maps of the teacher network and the student network with the $L1$ loss of the regression map. This allows the lightweight student network to have the ability to extract the spatial and spectral features of the deep teacher network, while significantly improving the computational efficiency. A deep learning inference optimizer is applied to further optimize the network by fusing the similar layers and quantifying the parameters, to fit the hardware environment and improve the inference efficiency. What is more, as shown in Fig. 1, a snapshot hyperspectral imager is connected to the hardware platform integrating the lightweight hyperspectral object tracking method to realize the real-time acquisition and onboard processing of hyperspectral video images. A hyperspectral object tracking system for edge applications is finally established. The major innovations of this article are summarized as follows:

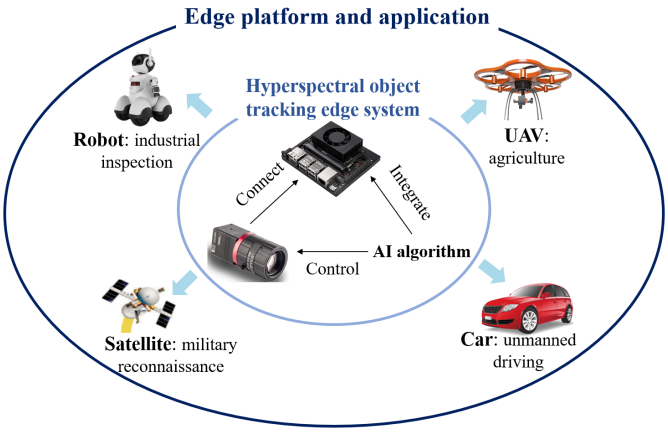


Fig. 1. Composition and applications of hyperspectral object tracking edge systems.

1) An embedded and edge platform-oriented lightweight dual Siamese network is proposed for hyperspectral object tracking to reduce the computational burden of hyperspectral data and network parameters.
2) The JSSKD method is proposed for compressing the parameters of the hyperspectral object tracking network while ensuring a minimal accuracy loss. Compared with the most efficient network at the present time, the inference speed is increased by 1.5 times, meeting the real-time requirement.
3) A prototype system for hyperspectral object tracking methods on embedded and edge platforms is introduced, which integrates a snapshot hyperspectral camera, an edge device, and the optimized lightweight hyperspectral object tracking model, to realize the function of real-time acquisition and tracking.

II. Related Work

This section introduces the classical and advanced methods for hyperspectral video object tracking. In addition, the methodological system of knowledge distillation and the knowledge distillation methods in object tracking are also presented.

A. Hyperspectral Video Object Tracking Methods

Owing to the rich spectral information, hyperspectral imagery can allow more precise classification [21], [22], [23], detection [24], [25], and tracking [1], [15] of the object of interest, achieving a better performance than previous methods [26], [27] on tiny and camouflaged objects. Hyperspectral video object tracking methods can be mainly classified into traditional methods, correlation filter based methods, and deep learning based methods, as with the RGB-based object tracking methods [28].

1) Traditional Methods: These methods generate several candidate frames at the object position in the previous frame. A manually designed feature extractor is then used to obtain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

the candidate frame features and match the candidate frames with traditional methods, such as the spectral angle mapper [29] or mean shift [30]. For example, Banerjee et al. [1] designed a method based on the spectral angle mapper to find the candidate frame with the smallest spectral difference as the tracking result. Nguyer et al. [8] designed features based on spectral reflectance and used a mean shift method which iterates to the extreme value point of the similarity function to enhance the performance of the model in some challenging scenarios such as occlusions and illumination changes.

2) Correlation Filter-Based Methods: These methods generate a filter template based on the location of the bounding box in the first hyperspectral image and then perform a correlation operation in the subsequent hyperspectral images using the feature map of the search region to find the position with the highest similarity as the center of the bounding box. Most of these methods concentrate on extracting hyperspectral features. For example, by utilizing a convolutional neural network to extract spatial–spectral features and then inputting the feature map into a classical correlation filter tracker [9], [10] or manually designing a hyperspectral feature extractor to obtain the material, structure, and other features of the hyperspectral images [11].

3) Deep Learning Based Methods: These methods use a large amount of data to train the deep network so that it learns the generic spatial–spectral representation and enhances the network feature extraction capability, which in turn enables the tracking of arbitrary targets. However, due to the limited training samples of hyperspectral video, it is difficult for a deep network to learn a generic representation. To solve this problem, most deep learning methods focus on the design of the band selection module, and convert the hyperspectral images into pseudo-color images with a high-performance RGB-based tracker [12], [13], [14]. However, these methods either employ complex feature extractors in expressing the hyperspectral features or have a large number of redundant computations in the network structure, resulting in less efficient algorithms that are far below real-time standards.

B. Knowledge Distillation

In order to improve the inference efficiency of deep learning-based networks, lightweight network structure design [31], [32] and model compression [33], [34] have attracted a lot of research interest in recent years. Knowledge distillation, which was first proposed by Hinton et al. [33], is widely used for model compression. Knowledge distillation involves employing a pretrained deep teacher network to guide the shallow student network training. The output of the student network is designed to approximate the teacher network by distillation loss, so as to transfer the knowledge between the teacher network and the student network and obtain a lightweight network with a similar accuracy. In terms of the type of knowledge transferred, the subsequent knowledge distillation tasks can be divided into response-based knowledge distillation, feature-based knowledge distillation, and relationship-based knowledge distillation [34].

1) Response-Based Knowledge Distillation: This involves using the unnormalized response values of the output layer neurons as soft labels and designing the loss function to calculate the difference between the soft labels of the teacher network and the student network, which in turn enables knowledge transfer at the output level.

2) Feature-Based Knowledge Distillation: This involves using the feature activation of the teacher network and the student network to calculate the loss function [35]. Compared with the response-based distillation approach, the feature-based knowledge distillation method can fully express the information of the hidden layer in the deep network. On this basis, certain modules and mechanisms have been employed to improve the performance of the matching of features, such as an attention mechanism [36], [37], probability distribution [38], and route-constrained hint learning [39].

3) Relationship-Based Knowledge Distillation: These methods transfer the knowledge from the correlation between different layers and samples. The knowledge is expressed by either designing a graph [40], [41], [42], a matrix [43], or a feature embedding [44], to represent the relationships between the feature maps or the outputs from several samples.

The knowledge distillation framework is applicable to numerous tasks [45], [46], [48], [49], including object tracking. The feature-based knowledge distillation approach was first applied by Zhu et al. [50], with the loss function used to evaluate the difference of the feature maps from the backbone VGG network in the teacher network and student network. Other methods have focused more on improving the knowledge distillation strategy. For example, reinforcement learning [51], as an approach to improve the optimization strategy, has been applied to distill the backbone network [52]. The teacher-student knowledge distillation strategy was proposed to enable in-depth knowledge understanding, and consists of a single teacher-student knowledge distillation module and a multistudent knowledge sharing mechanism [53]. Feature maps from the template image and search image have also been matched to calculate a correlation tracking loss that estimates the discriminative ability of the student network [54]. As for the response-based knowledge distillation methods, the distillation loss has been formulated as the weighted sum of the Kullback–Leibler (KL) divergence of the classification map and the $L1$ loss of the regression map [55].

III. Hyperspectral Object Tracking Model Compression and Deployment

In this section, a complete pipeline for the lightweight hyperspectral object tracking method is presented, as shown in Fig. 2, consisting of three main steps: 1) the dual Siamese network architecture; 2) joint spatial–spectral knowledge distillation; and 3) compressed model deployment. The dual Siamese network—SiamHYPER [15]—introduces a pretrained RGB-based model to enhance the feature extraction capability of the model and alleviate the data-hungry problem caused by the insufficient samples of hyperspectral video. However, the computational overhead caused by twice the number of parameters makes it difficult for the network to reach the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

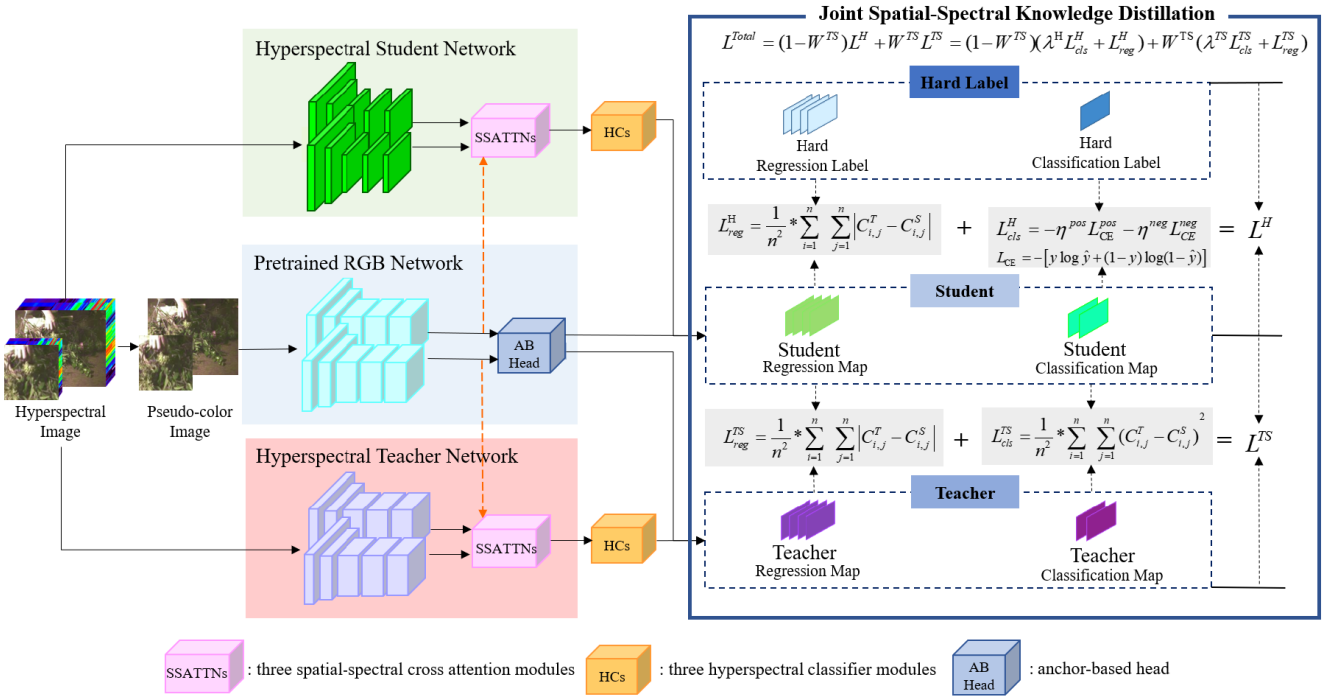


Fig. 2. Joint spatial–spectral knowledge distillation for hyperspectral object tracking model compression based on a dual Siamese network.

speed of real-time processing. Therefore, a spatial–spectral joint distillation method for the dual Siamese network is proposed to achieve the reduction of model computation and to realize the onboard processing of hyperspectral object tracking models on embedded edge devices with inference optimization techniques.

A. Dual Siamese Network Architecture

SiamHYPER [15] is based on a dual Siamese network architecture consisting of an RGB-based tracker and a hyperspectral tracker. The RGB-based tracker is pretrained with a large amount of RGB data to learn the general representation, which alleviates the data-hungry problem caused by the insufficient sample size and allows the model to fit more appropriately. The parameters of the RGB-based tracker are frozen during training. The spatial–spectral cross-attention (SSATTN) module and hyperspectral classifier module are proposed for feature fusion and aggregation. Specifically, the feature maps from the last three blocks of the hyperspectral backbone are input into three SSATTN modules and fused with the feature maps from the same hierarchy of the RGB backbone, so as to use the pretrained RGB tracker to guide the hyperspectral tracker in learning the spatial features of the hyperspectral images. After this, three cascaded hyperspectral classifier modules aggregate the hierarchical feature maps from the SSATTN modules and produce the classification map and regression map. SiamHYPER [15] uses a classical Siamese network architecture with the backbone of ResNet50 in expressing hyperspectral features, and the speed can be close to the real-time tracking standard. The introduction of pretraining parameters ensures the basic feature extraction capability of the network, allowing the accuracy of the model to surpass that of the previous methods.

B. Joint Spatial-Spectral Knowledge Distillation

A joint spatial–spectral knowledge distillation framework is proposed for compressing the SiamHYPER [15] dual Siamese network. The original hyperspectral Siamese network is chosen as the teacher network, with the parameters frozen during distillation, while the hyperspectral Siamese network with a lightweight backbone is the student network.

1) Distillation Architecture: A lightweight student network with the backbone of ResNet18 is first chosen. Due to the rich spatial–spectral information of hyperspectral images, a deeper backbone network is usually required to extract hyperspectral features, which brings more computational costs. With the guidance of a teacher network, a shallower network can approximate the feature extraction capability of the teacher network, thus reducing the computational burden of the model. Therefore, the ResNet18 backbone, as a lightweight backbone, was chosen to replace the original ResNet50 and is used in SiamOHOT for the in-depth expression of hyperspectral features. The soft labels obtained from the classification map and regression map are used for the distillation, because of the rich information of the positive and negative samples, to enhance the discriminative ability of the network. In the framework of SiamHYPER [15], the RGB tracker is responsible for extracting spatial features from the three-band pseudo-color images, while the hyperspectral tracker needs to extract both spatial and spectral features from the hyperspectral images. The SSATTN module enhances the semantic representation, but the hyperspectral tracker is still inadequate for the spatial features. By using only the hyperspectral tracker as a teacher network, the spatial feature extraction capability of the distilled obtained student network will be further degraded, which in turn weakens the discriminative power of the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

Therefore, the classification map output by the RGB tracker is introduced to guide the student network, together with the classification map output by the hyperspectral tracker of the teacher network. The classification maps output by the hyperspectral tracker of the teacher network and the hyperspectral tracker of the student network are fused with the classification map output by the RGB tracker. The loss of the fused classification maps of the student network and the teacher network is then calculated, allowing the student network to learn both the powerful spatial feature extraction capability of the pretrained RGB tracker and the spectral information extraction capability of the hyperspectral tracker, thus enhancing the representation of spatial–spectral information by the student network.

2) Distillation Loss: The loss function of JSSKD includes hard loss and teacher-student loss. Specifically, the hard loss is calculated from the output of the student network and the ground truth, while the teacher-student loss is used to penalize the differences between the outputs of the teacher network and the student network.

When calculating the teacher-student loss, the soft labels obtained from the classification map and the regression map of the teacher network and student network are utilized. Unlike the hard labels, the distribution of values in the soft labels is more discrete and contains more useful information. For example, each value in the classification map represents the probability of the anchor at that location being a positive or negative sample, and each value in the regression plot represents the exact coordinates of the bounding box at that location. The loss of the classification and regression plots of the teacher-student network allows the student network to learn more information about the teacher network, which in turn allows the convergence position of the student network to approach that of the teacher network. Specifically, the mse loss function is used for calculating the loss of the teacher-student classification maps, which measures the classification score of each anchor on the categorical graph. The formulation of the teacher-student classification loss $L_{\text{cls}}^{\text{TS}}$ is as follows:

$$L_{\text{cls}}^{\text{TS}} = \frac{1}{n^2} * \sum_{i=1}^{n} \sum_{j=1}^{n} \left(C_{i,j}^{T} - C_{i,j}^{S}\right)^2 \tag{1}$$

where n denotes the size of the classification map, and $C_{i,j}^{T}$ and $C_{i,j}^{S}$ denote the value of position (i, j) on the classification map of the teacher network and student network, respectively. The differences between the regression maps of the teacher network and student network are quantized by $L1$ loss. $L1$ loss, which is also named the mean absolute error (MAE), describes the sum of the absolute differences between the target and predictor variables and is used to measure the average size of the error in a set of predicted values. The teacher-student regression loss $L_{\text{reg}}^{\text{TS}}$ can be expressed as

$$L_{\text{reg}}^{\text{TS}} = \frac{1}{n^2} * \sum_{i=1}^{n} \sum_{j=1}^{n} \left|C_{i,j}^{T} - C_{i,j}^{S}\right|. \tag{2}$$

Hard loss is widely used in the training process of deep networks to make the output of the network approximate the ground truth. It is also vital to forbid the transmission of wrong knowledge when the tracking of the teacher network fails in some difficult scenes. The hard regression loss L_{reg}^{H} is the same as $L_{\text{reg}}^{\text{TS}}$, while the cross-entropy loss function is applied to calculate the classification hard loss L_{cls}^{H}. The cross-entropy loss L_{CE} can effectively avoid the disadvantage of the slow gradient update of functions similar to the sigmoid form, and prevent gradient dispersion, and thus convergence speed reduction during the gradient descent. The cross-entropy loss L_{CE} is formulated as

$$L_{\text{CE}} = -\left[y \log \hat{y} + (1 - y) \log(1 - \hat{y})\right] \tag{3}$$

where y and $\hat{y}$ are 1-D vectors stretched from the student network classification map and hard labels. In order to effectively balance the influence of positive and negative samples, we separate positive and negative samples from the hard labels and classification map based on the distribution of 0 and 1 in the hard labels, and calculate the classification hard loss

$$L_{\text{cls}}^{H} = -\eta^{\text{pos}} L_{\text{CE}}^{\text{pos}} - \eta^{\text{neg}} L_{\text{CE}}^{\text{neg}} \tag{4}$$

where η^{pos} and η^{neg} are coefficients of the loss of the positive and negative samples, $L_{\text{CE}}^{\text{pos}}$ is the cross-entropy loss of the classification map and hard labels, and $L_{\text{CE}}^{\text{neg}}$ is the negative samples.

The complete loss function L^{Total} is formulated as follows:

$$L^{\text{Total}} = (1 - W^{\text{TS}})\left(\lambda^{H} L_{\text{cls}}^{H} + L_{\text{reg}}^{H}\right) + W^{\text{TS}}\left(\lambda^{\text{TS}} L_{\text{cls}}^{\text{TS}} + L_{\text{reg}}^{\text{TS}}\right) \tag{5}$$

where W^{TS}, λ^{H}, and λ^{TS} are the coefficients. Specifically, the classification map determines the center of the bounding box, which directly determines the success of the tracking and is more important than the regression map that fine-tunes the position of the enclosing frame. Therefore, the values of λ^{H} and λ^{TS} are usually greater than 1 to improve the discriminative property of the network.

C. Compressed Model Deployment

To realize the online processing of the model in the edge environment, the lightweight model is deployed on an edge device, and the model is further optimized using a deep learning inference optimizer, to realize the complete process from acquisition, processing, and to result storage in the edge environment.

A real-time data stream from the acquisition side to the processing side is established by connecting the snapshot hyperspectral camera to the edge device through a hardware and software interface. A 16-channel hyperspectral video camera (XIMEA, IMEC, Belgium) with a spectral range from 400 to 600 nm is used on the acquisition side. The NVIDIA Jetson Xavier NX, with a memory of 16 GB, was chosen as the edge device in this study. The development board has a computational power of 21 trillions of operations per second (TOPS) at 20 W to support deep learning model inference. In terms of the hardware interface, the hyperspectral video camera is connected to the development board using a USB interface. As for the software, the xiAPI toolkit is installed as the software interface to control the camera. Through the development of the software API, the essential

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

functions such as acquisition, saving, and the setting of the frame rate and exposure time can be realized. Based on this, the front-end interface is built based on the Python PyQT plug-in, which integrates the functions of real-time acquisition and display of hyperspectral images, key frame interception, object of interest labeling, and real-time tracking by calling the tracking model.

To adapt to the hardware and software resources, the TensorRT deep learning inference optimizer is used to further optimize the model, simplify the network architecture, and improve the efficiency of the underlying hardware resource invocation. During the deep learning model inference, the convolutional layer, activation layer, and normalization layer call the corresponding software interface of the graphics processing unit (GPU) separately. After the layer and tensor fusion, the original network layers are combined into one module, which can reduce the number of GPU calls and improve the operation speed of the network while ensuring that the basic functions of the network remain unchanged. The data type of a deep model is usually a 32-bit floating number during the inference, where the longer data bits make the data more accurate while occupying more storage and computational resources. The precision calibration technique can realize a lightweight data type, reducing the 32-bit floating point numbers to 16-bit floating point numbers or 8-bit integers, which reduces the resource occupation while minimizing the precision loss and facilitating the deployment in the onboard environment. The optimized efficient lightweight network can be deployed on an embedded edge computing development board to realize real-time efficient processing of hyperspectral object tracking models.

IV. Experiments and Results

A. Implementation Details

The proposed method was implemented in Python with PyTorch 1.12.0, and the hardware platform consisted of an Intel Xeon Gold 5218R CPU, RAM of 125 GB, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. SiamRPN++ was chosen as the RGB Siamese network, which was pretrained on the COCO [56], ImageNet DET [57], ImageNet VID [57], and YouTube-BoundingBoxes [58] datasets. The first, ninth, and 16 bands were selected to constitute the three-band images as the input of the RGB Siamese network. The whole hyperspectral image was chosen as the input of the hyperspectral Siamese network. The classification maps output by the hyperspectral tracker of the teacher network and the hyperspectral tracker of the student network were fused with the classification map output by the RGB tracker to enrich the semantic information. The optimizer was stochastic gradient descent, with a weight decay and momentum of 0.0001 and 0.9, respectively, and a warmup learning rate from 0.001 to 0.005. The training epochs were set to 20, while 20 000 samples were generated in each epoch, with the batch size of 30. The weights of the distillation loss and tracking loss were set to 0.9 and 0.1, respectively.

B. Dataset and Evaluation Metrics

The proposed method was tested on the 2021 WHISPERS Hyperspectral Object Tracking Challenge dataset. The test set contains 35 video sequences with a total of about 16 500 frames, and the training set contains 40 video sequences with a total of about 12 800 frames. Each video sequence includes hyperspectral video data, synthetic three-band pseudo-color data, and RGB video data, which are aligned with the hyperspectral video images. To fully evaluate the performance of a hyperspectral object tracking model, the dataset is designed with scenes containing various challenging attributes, such as occlusions, rotation, fast motion, deformation, etc. Specifically, in order to reflect the vital influence of hyperspectral images, the dataset is designed with a large number of background scenes, such as a person dressed in black walking into shadows, red cans flipping and rolling over a red cloth, and green fruits moving among tree branches and leaves. The foreground and background are similar in terms of color and texture, requiring the model to make full use of the spectral information to distinguish the objects.

The one-pass evaluation benchmark was chosen to evaluate the performance of the methods, evaluating the area under the curve (AUC) score of the success plot and the average distance precision (DP) score of the precision plot. The success plot shows the overlap ratio of the predicted bounding box and ground truth. With the threshold varied from 0 to 1 as the horizontal axis of the success plot and the percentage of the frames with an overlap ratio higher than the threshold as the vertical axis, the larger the AUC score of the success plot the better. When the Euclidean distance between the center of the predicted bounding box of a frame and the ground truth is less than the threshold, the tracking result of this frame is considered precise. The precision plot, similar to the success plot, shows the percentage of precise frames under different thresholds varied from 0 to 50. Specifically, the DP score is the precision rate with a threshold of 20.

C. Comparison With the State-of-the-Art

In order to validate the effectiveness of the proposed method, several hyperspectral trackers were chosen for comparison, i.e., the SiamHYPER [15] teacher network, a correlation filter-based method (MHT) [11], a convolutional network-based hyperspectral tracking method (CNHT) [9], and a deep learning based method (BAE-Net) [12]. Some high-performance RGB trackers, i.e., SiamRPN++ [59], ATOM [60], KeepTrack [61], and ToMP [62], pretrained on RGB datasets were also selected and tested on the pseudo-color subsets of the dataset. Specifically, SiamRPN++ [59] and ToMP [62] used the backbone of ResNet50, as with SiamHYPER [15].

As shown in Fig. 3, compared with the SiamHYPER [15] teacher network, the AUC score and DP score of the SiamOHOT student network decrease by 0.05 and 0.056, respectively. The backbone of the SiamOHOT student network is shallower, and the ability to extract the spectral information of hyperspectral images is slightly weaker than that of the SiamHYPER [15] teacher network. Compared with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

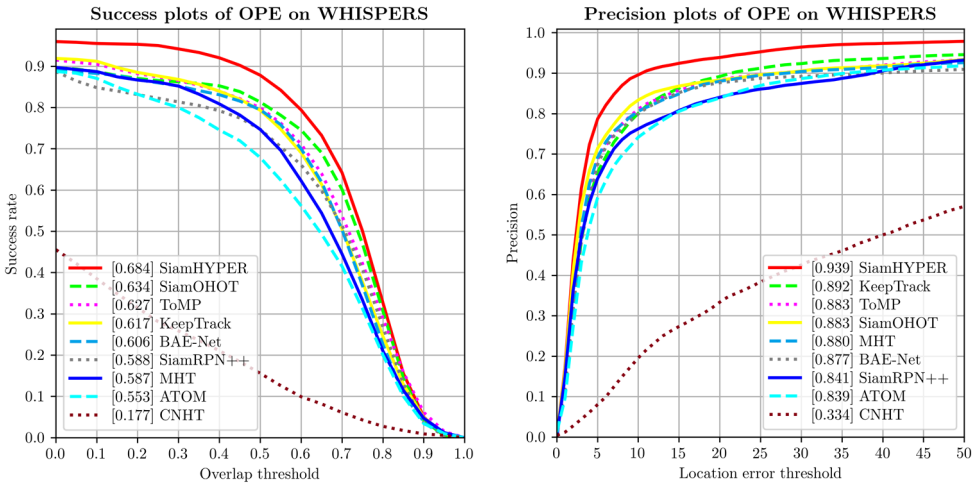


Fig. 3. Comparison of the hyperspectral trackers on hyperspectral videos and the RGB trackers on pseudo-color videos. (a) Success plot. (b) Precision plot.

other hyperspectral trackers and RGB trackers, SiamOHOT achieves the leading accuracy in the success plot, exceeding the third-best method of ToMP [62] by 0.007 and exceeding BAE-Net [12] by 0.028. SiamOHOT, as a dual Siamese network architecture, retains the network structure and pretraining parameters of the RGB tracker and is still stronger than the other methods. In terms of the DP score, which mainly measures the localization accuracy of the algorithm, as the third-best method, the SiamOHOT student network achieves a lower score than the high-performance RGB tracker KeepTrack [61] by 0.009, and has the same DP score as ToMP. The shallow network makes the student network less accurate in some scenes, but it also achieves high precision and better performance than the other hyperspectral trackers.

In terms of efficiency, the number of parameters for the hyperspectral branch of SiamOHOT is 101 MB, which is reduced by more than half when compared to SiamHYPER [15] with 225 MB. As shown in Table I, the inference speed of SiamOHOT is improved by 13 frames/s when compared to the SiamHYPER [15] teacher network, owing to the lighter network structure design of the hyperspectral branch. In contrast, because the Siamese network structure relies on a large-scale pretrained RGB tracker to guide the hyperspectral tracker training, its network structure and parameters cannot be changed, to ensure the accuracy of the RGB tracker. The number of parameters for the RGB branch is difficult to compress sufficiently, which limits the further improvement of the Siamese network inference efficiency. Compared with the other hyperspectral trackers, the inference speed of SiamOHOT can reach ten times that of the other methods. Among the other methods, MHT [11] uses a spectral-spatial histogram of multidimensional gradients and fractional abundances to express the hyperspectral features, which brings a huge computational burden. BAE-Net [12] generates multiple three-channel images from hyperspectral images for input into the RGB tracker, repeating the inference process multiple times and doubling the processing time as a result. CNHT [9] uses 3-D patches generated from the target region of the hyperspectral image as convolution kernels. Compared with the traditional 2-D convolution operation, this also brings an excessive computational burden. SiamOHOT, distilled from the SiamHYPER [15] teacher network, combines the advanced model of the dual Siamese network framework with a more efficient feature approach than manually designed features. As a result, it avoids the multiple computations of the three-band pseudo-color image tracking methods, and features a more lightweight network structure, making it the first hyperspectral tracking algorithm to meet the real-time standard. The inference speeds of the RGB trackers are not mentioned because of the different inputs.

TABLE I
AUC/DP SCORES AND FPS OF THE HYPERSPECTRAL TRACKERS AND RGB TRACKERS

Tracker	AUC/DP	FPS	Hardware
SiamRPN++ [59]	0.588/0.841	/	NVIDIA GeForce RTX 3090 GPU
ATOM [60]	0.553/0.839	/	NVIDIA GeForce RTX 3090 GPU
KeepTrack [61]	0.617/0.892	/	NVIDIA GeForce RTX 3090 GPU
ToMP [62]	0.627/0.883	/	NVIDIA GeForce RTX 3090 GPU
MHT [11]	0.587/0.880	0.5	Intel Core i7 2.8 GHz CPU
BAE-Net [12]	0.606/0.877	0.5	NVIDIA TITAN X (Pascal) GPU
CNHT [9]	0.177/0.334	0.8	Intel Core i7 2.8 GHz CPU
SiamHYPER [15]	0.684/0.939	25	NVIDIA GeForce RTX 3090 GPU
SiamOHOT	0.634/0.883	38	NVIDIA GeForce RTX 3090 GPU

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

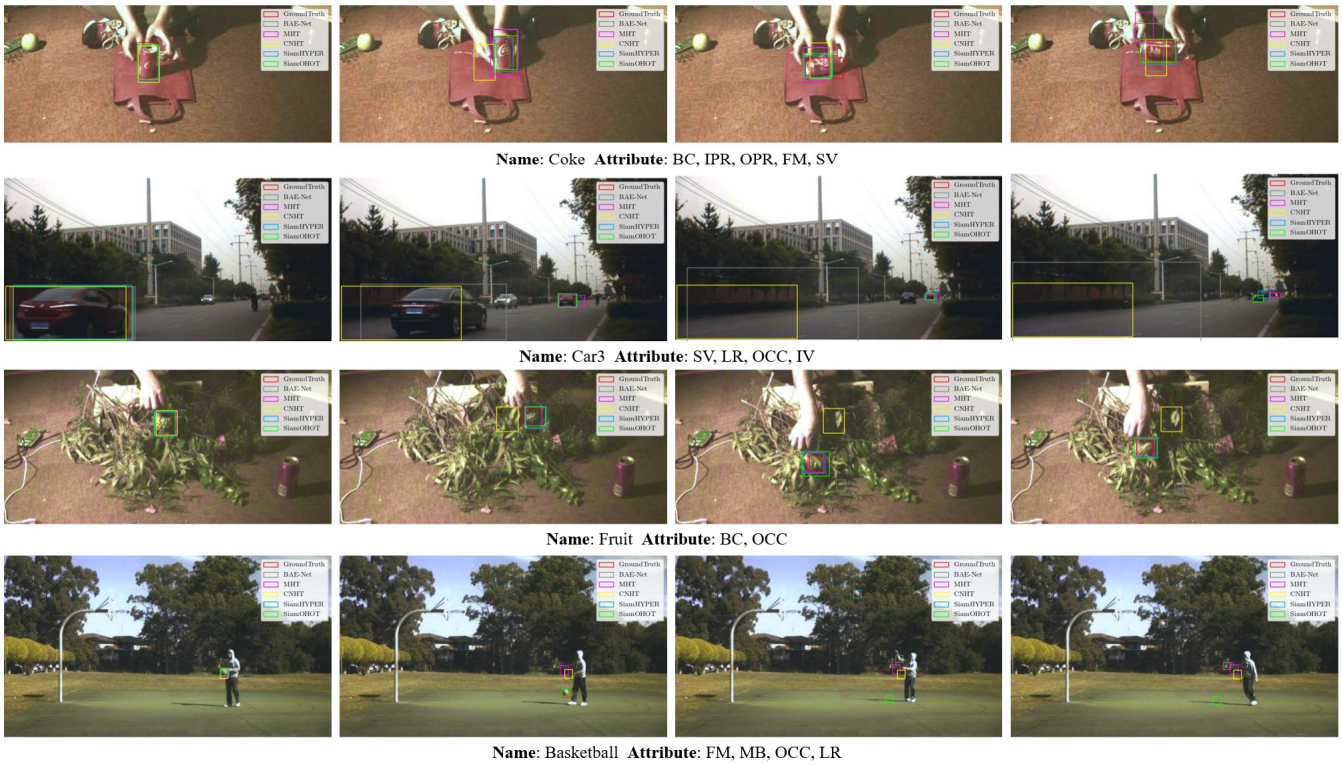


Fig. 4. Qualitative evaluation of some representative video sequences.

TABLE II
ATTRIBUTE-BASED COMPARISON

Attribute	SiamRPN++ [59]	ATOM [60]	KeepTrack [61]	ToMP [62]	BAE-Net [12]	MHT [11]	CNHT [9]	SiamHYPER [15]	SiamOHOT
BC	0.556	0.573	0.605	0.577	0.631	0.594	0.183	0.722	0.685
DEF	0.689	0.639	0.701	0.738	0.679	0.664	0.294	0.711	0.713
FM	0.656	0.506	0.638	0.692	0.607	0.541	0.189	0.702	0.560
IPR	0.645	0.618	0.691	0.712	0.699	0.670	0.278	0.744	0.730
IV	0.480	0.510	0.564	0.566	0.439	0.473	0.087	0.602	0.504
LR	0.529	0.473	0.565	0.621	0.491	0.478	0.029	0.659	0.498
MB	0.634	0.531	0.683	0.698	0.594	0.560	0.102	0.721	0.607
OCC	0.565	0.543	0.615	0.589	0.555	0.565	0.123	0.637	0.556
OPR	0.686	0.624	0.686	0.700	0.693	0.631	0.265	0.728	0.722
OV	0.606	0.558	0.626	0.715	0.515	0.620	0.149	0.599	0.579
SC	0.537	0.692	0.713	0.703	0.666	0.646	0.038	0.773	0.732
SV	0.579	0.524	0.600	0.640	0.608	0.563	0.152	0.661	0.633

The red font indicates the best result, and the green font indicates the second-best result. The attributes are background clutter (BC), deformation (DEF), fast motion (FM), in-plane rotation (IPR), illumination variation (IV), low resolution (LR), motion blur (MB), occlusion (OCC), out-of-plane rotation (OPR), out-of-view (OV), scale change (SC), and scale variation (SV).

As shown in Fig. 4, on some representative sequences such as the coke, car3, and fruit sequences, SiamOHOT achieves an outstanding performance, as does SiamHYPER [15]. Compared with the other hyperspectral trackers, the qualitative performance is significantly better. However, in challenging scenes such as fast motion and low resolution, SiamOHOT is not robust enough when compared with the teacher network, because of the fact that the shallower network has a weaker feature expression ability and a smaller receptive field. When the target is tiny or moving too fast, SiamOHOT is more likely to suffer from tracking drift.

In order to further explore the performance limit of the SiamOHOT student network in different scenes, a comparison of the AUC scores under the scenes with different attributes was conducted. From the attribute-based results listed in Table II, in nearly half of the scenarios, the AUC score of the SiamOHOT student network is second only to that of the SiamHYPER [15] teacher network, and is better than the other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EXPERIMENT SHOWING THE EFFECTIVENESS OF KNOWLEDGE DISTILLATION

	SiamHYPER [15]	SiamOHOT	SiamOHOT-lfs
AUC	0.684	0.634	0.626
DP	0.939	0.883	0.868

TABLE IV
EXPERIMENT SHOWING THE EFFECTIVENESS OF THE RGB TRACKER IN JSSKD

	SiamHYPER [15]	SiamOHOT	SiamOHOT-no_RGB
AUC	0.684	0.634	0.623
DP	0.939	0.883	0.864

TABLE V
EXPERIMENT SHOWING THE EFFECTIVENESS OF THE HYPERPARAMETERS IN JSSKD

	W^{TS}	λ^{H} (λ^{TS})	**AUC**	**DP**
1	0.9	2	0.634	0.883
2	0.9	1	0.632	0.872
3	0.9	0.5	0.631	0.879
4	0.5	2	0.631	0.874
5	0.5	1	0.632	0.877
6	0.5	0.5	0.633	0.880
7	0.1	2	0.629	0.868
8	0.1	1	0.633	0.876
9	0.1	0.5	0.633	0.873

TABLE VI
ACCURACY AND EFFICIENCY OF THE HYPERSPECTRAL TRACKERS DEPLOYED ON THE EDGE PLATFORM

	AUC	**DP**	**FPS**	**Optimized**
MHT [11]	0.587	0.880	<0.1	×
BAE-Net [12]	0.606	0.877	<0.1	×
CNHT [9]	0.177	0.334	<0.1	×
SiamHYPER [15]	0.684	0.939	1.7	×
SiamOHOT	0.634	0.883	2.4	×
			3.2	√

hyperspectral trackers. In the deformation scenario, it even achieves a better performance than the SiamHYPER [15] teacher network. In the background clutter scene, SiamOHOT achieves a much better performance than the RGB trackers when making full use of the spectral information. In the out-of-view scene, the target truck comes from far and near, the initial frame contains only the front end, and the middle frame gradually appears. Both the teacher and the student network focus only on the front end of the truck. Generally speaking, SiamOHOT performs better than the other hyperspectral trackers and almost all the RGB trackers in most scenes, but the lightweight structure of the SiamOHOT student network makes it difficult to express hyperspectral features effectively when both the spatial and spectral information of the target produces large changes, resulting in accuracy degradation. Where spectral information contributes less in some scenes, such as the deformation and fast motion scenes, SiamOHOT performs slightly worse than the state-of-the-art RGB trackers.

D. Ablation Study

Three ablation experiments were designed to prove the effectiveness of the knowledge distillation, the effectiveness of the classification map of the RGB tracker in JSSKD, and the setting of the hyperparameters in (5).

In the first ablation experiment, a student network learning from scratch (SiamOHOT-lfs) was introduced and compared with the proposed SiamOHOT using JSSKD during training. As shown in Table III, compared to the student network not trained with knowledge distillation, the AUC score and DP score of SiamOHOT are improved by 0.008 and 0.015, respectively, demonstrating the facilitative effect of knowledge from the teacher network on the student network. Using the supervision information of the teacher network allows the student network to be close to the convergence space of the teacher network, obtaining the characteristic expressiveness of the deep teacher network more quickly and accurately.

In the second ablation experiment, the classification map from the RGB tracker was removed from the knowledge distillation strategy, producing student network training without spatial information from RGB (SiamOHOT-no_RGB). The results in Table IV show that the AUC score and DP score decline by 0.011 and 0.019, respectively. The RGB tracker, pretrained on large-scale RGB data, has a strong expressive power for spatial representations and can complement the weakness of the hyperspectral teacher network in spatial feature extraction ability. Therefore, utilizing the output of the RGB tracker does improve the feature expression capability of the SiamOHOT student network.

In the third ablation experiment, we tried different values for the hyperparameters in (5). Specifically, the weight of the teacher-student loss W^{TS} was set to 0.9, 0.5, and 0.1. λ^{H} was set to 2, 1, and 0.5, demonstrating the weight of the classification map loss, which was the same as λ^{TS}. Table V shows that SiamOHOT achieves the best performance when W^{TS}, λ^{H}, and λ^{TS} are set to 0.9, 2, and 2, respectively. This proves that the distillation loss is more important than the hard loss in knowledge distillation. It also indicates that the classification map containing the rich semantic information of the RGB and hyperspectral features should have a higher weight.

E. Performance on the Prototype Edge System

A prototype system consisting of the tracking algorithm, a snapshot hyperspectral sensor, and an NVIDIA Jetson Xavier NX platform was established to evaluate the onboard

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

performance of the hyperspectral trackers. As shown in Table VI, when processing the real-time data stream from the snapshot hyperspectral sensor, SiamHYPER [15] can only infer at the speed of 1.7 frames/s, while MHT [11], BAE-Net [12], and CNHT [9] work at less than 0.1 frames/s. SiamOHOT, with the lightweight structure, improves the speed to 2.4 frames/s. What is more, after further optimization with the TensorRT deep learning inference optimizer, the optimized SiamOHOT network finally achieves a speed of 3.2 frames/s, which is nearly double that of SiamHYPER [15], demonstrating the great improvement of the proposed method for the real-time processing of hyperspectral video object tracking on an edge platform.

V. Conclusion and Future Work

To address the problems of the huge computational cost and slow inference speed of the existing hyperspectral tracking models and the conflict between the huge amount of hyperspectral video data and the insufficient storage resources, a joint spatial–spectral knowledge distillation method has been proposed in this article (SiamOHOT). A dual Siamese hyperspectral network with a lightweight backbone network was chosen as the student model, while the SiamHYPER [15] tracker was selected as the teacher model. The SiamOHOT student network can simultaneously learn the spatial feature extraction capability of the RGB model and the spectral feature expression capability of the hyperspectral model, thus improving the inference efficiency to meet the real-time requirements with only a slight loss of accuracy. The lightweight SiamOHOT network is then optimized further using the deep learning inference optimizer technique and is deployed on the edge device. The optimized SiamOHOT network occupies less memory resources and has fewer system resource calls, so that more resources are saved when processing huge amounts of hyperspectral video data. Finally, the edge device is connected to a snapshot hyperspectral imager, which acquires hyperspectral video images in real time, and the hyperspectral object tracking algorithm then processes these images in real time, realizing the whole process of onboard processing.

The work described in this article could be further improved in some aspects, which will be the direction of our future work. First, in the design of the SiamOHOT student network, the compression rate of the student network is not high enough because the structure of the Siamese network is not easy to change, which leads to a limited speed-up effect after knowledge distillation. We will attempt to adopt a lighter network architecture for the pretraining stage of the RGB model, and will combine this with neural network search technology to build a lighter student network. In terms of deep learning model deployment, it is difficult to directly optimize the whole object tracking network using TensorRT, due to the limitation of the operator and network structure. Only the computationally intensive part of the distilled network is optimized, and the speed of the onboard processing could be further improved. In the future, we also plan to try various embedded hardware platforms to find better onboard processing solutions.

References

[1] A. Banerjee, P. Burlina, and J. Broadwater, "Hyperspectral video for illumination-invariant tracking," in *Proc. 1st Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, 2009, pp. 1–4.

[2] A. Lambrechts, P. Gonzalez, B. Geelen, P. Soussan, K. Tack, and M. Jayapala, "A CMOS-compatible, integrated approach to hyper- and multispectral imaging," in *IEDM Tech. Dig.*, Dec. 2014, pp. 10.5.1–10.5.4.

[3] A. Bodkin, A. Sheinis, A. Norton, J. Daly, S. Beaven, and J. Weinheimer, "Snapshot hyperspectral imaging: The hyperpixel array camera," in *Proc. SPIE*, vol. 2009, Apr. 2009, Art. no. 73340H.

[4] Z. Xiong, L. Wang, H. Li, D. Liu, and F. Wu, "Snapshot hyperspectral light field imaging," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6873–6881.

[5] Y. Zhong et al., "Mini-UAV-Borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[6] A. Mikołajczyk and M. Grochowski, "Data augmentation for improving deep learning in image classification problem," in *Proc. Int. Interdiscipl. PhD Workshop (IIPhDW)*, May 2018, pp. 117–122.

[7] G. Marcus, "Deep learning: A critical appraisal," 2018, *arXiv:1801.00631*.

[8] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.-Workshops*, Jun. 2010, pp. 44–51.

[9] K. Qian, J. Zhou, and F. Xiong, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Smart Multimedia*. Cham, Switzerland: Springer, 2018, pp. 308–319.

[10] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[11] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[12] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2106–2110.

[13] Z. Li, X. Ye, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Spectral-Spatial-Temporal attention network for hyperspectral tracking," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[14] Y. Tang, Y. Liu, L. Ji, and H. Huang, "Robust hyperspectral object tracking by exploiting background-aware spectral information with band selection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] Z. Liu, X. Wang, Y. Zhong, M. Shu, and C. Sun, "SiamHYPER: Learning a hyperspectral object tracker from an RGB-based tracker," *IEEE Trans. Image Process.*, vol. 31, pp. 7116–7129, 2022.

[16] C. Zhao, H. Liu, N. Su, L. Wang, and Y. Yan, "RANet: A reliability-guided aggregation network for hyperspectral and RGB fusion tracking," *Remote Sens.*, vol. 14, no. 12, p. 2765, Jun. 2022.

[17] C. Zhao, H. Liu, N. Su, and Y. Yan, "TFTN: A transformer-based fusion tracking framework of hyperspectral and RGB," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542515.

[18] J. Shao, B. Du, C. Wu, and L. Zhang, "Tracking objects from satellite videos: A velocity feature based correlation filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7860–7871, Oct. 2019.

[19] W. Song et al., "A joint Siamese attention-aware network for vehicle object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625617.

[20] J. Shao, B. Du, C. Wu, and L. Zhang, "Can we track targets from space? A hybrid kernel correlation filter tracker for satellite video," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8719–8731, Nov. 2019.

[21] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[22] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[23] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.

[24] H. Gao, Y. Zhang, Z. Chen, S. Xu, D. Hong, and B. Zhang, "A multidepth and multibranch network for hyperspectral target detection based on band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506818.
[25] J. Li, X. Wang, S. Wang, H. Zhao, L. Zhang, and Y. Zhong, "One-step detection paradigm for hyperspectral anomaly detection via spectral deviation relationship learning," 2023, *arXiv:2303.12342*.
[26] X. Wu, D. Hong, J. Tian, J. Chanussot, W. Li, and R. Tao, "ORSIm detector: A novel object detection framework in optical remote sensing imagery using spatial-frequency channel features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5146–5158, Jul. 2019.
[27] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.
[28] F. Chen, X. Wang, Y. Zhao, S. Lv, and X. Niu, "Visual object tracking: A survey," *Comput. Vis. Image Understand.*, vol. 222, Sep. 2022, Art. no. 103508.
[29] M. A. Cho, P. Debba, R. Mathieu, L. Naidoo, J. van Aardt, and G. P. Asner, "Improving discrimination of savanna tree species through a multiple-endmember spectral angle mapper approach: Canopy-level analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4133–4142, Nov. 2010.
[30] D. Comaniciu, V. Ramesh, and P. Meer, "Real-time tracking of non-rigid objects using mean shift," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2000, pp. 142–149.
[31] Y. Wan, Y. Zhong, A. Ma, J. Wang, and L. Zhang, "E2SCNet: Efficient multiobjective evolutionary automatic search for remote sensing image scene classification network architecture," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Nov. 17, 2022, doi: 10.1109/TNNLS.2022.3220699.
[32] A. Ma, Y. Wan, Y. Zhong, J. Wang, and L. Zhang, "SceneNet: Remote sensing scene classification deep learning network using multi-objective neural evolution architecture search," *ISPRS J. Photogramm. Remote Sens.*, vol. 172, pp. 171–188, Feb. 2021.
[33] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[34] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021.
[35] A. Romero, N. Ballas, S. Ebrahimi Kahou, A. Chassang, C. Gatta, and Y. Bengio, "FitNets: Hints for thin deep nets," 2014, *arXiv:1412.6550*.
[36] Z. Huang and N. Wang, "Like what you like: Knowledge distill via neuron selectivity transfer," 2017, *arXiv:1707.01219*.
[37] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," 2016, *arXiv:1612.03928*.
[38] N. Passalis and A. Tefas, "Learning deep representations with probabilistic knowledge transfer," in *Proc. Int. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2018, pp. 283–299.
[39] X. Jin et al., "Knowledge distillation via route constrained optimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1345–1354.
[40] C. Zhang and Y. Peng, "Better and faster: Knowledge transfer from multiple self-supervised learning tasks via graph distillation for video classification," 2018, *arXiv:1804.10069*.
[41] S. Lee and B. Cheol Song, "Graph-based knowledge distillation by multi-head attention network," 2019, *arXiv:1907.02226*.
[42] Y. Liu et al., "Knowledge distillation via instance relationship graph," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7089–7097.
[43] J. Yim, D. Joo, and J. Bae, "A gift from knowledge distillation: Fast optimization, network minimization and transfer learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2017, pp. 7130–7138.
[44] H. Chen, Y. Wang, C. Xu, C. Xu, and D. Tao, "Learning student networks via feature embedding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 25–35, Jan. 2021.
[45] Y. Hu, X. Huang, X. Luo, J. Han, X. Cao, and J. Zhang, "Variational self-distillation for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627313.
[46] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.
[47] Z. Duan, S. Wang, H. Di, and J. Deng, "Distillation remote sensing object counting via multi-scale context feature aggregation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613012.
[48] M. Zhou, J. Huang, X. Fu, F. Zhao, and D. Hong, "Effective pan-sharpening by multiscale invertible neural network and heterogeneous task distilling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411614.
[49] Y. Zhang, Z. Yan, X. Sun, W. Diao, K. Fu, and L. Wang, "Learning efficient and accurate detectors with dynamic knowledge distillation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613819.
[50] G. Zhu, J. Wang, P. Wang, Y. Wu, and H. Lu, "Feature distilled tracking," *IEEE Trans. Cybern.*, vol. 49, no. 2, pp. 440–452, Feb. 2019.
[51] M. Minsky, "Steps toward artificial intelligence," *Proc. IRE*, vol. 49, no. 1, pp. 8–30, Jan. 1961.
[52] M. Dunnhofer, N. Martinel, and C. Micheloni, "Tracking-by-trackers with a distilled and reinforced model," in *Proc. Asian Conf. Comput. Vis. (ACCV)*, Cham, Switzerland: Springer, 2021, pp. 631–650.
[53] J. Shen, Y. Liu, X. Dong, X. Lu, F. S. Khan, and S. Hoi, "Distilled Siamese networks for visual tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 8896–8909, Dec. 2022.
[54] Q. Chen, B. Zhong, Q. Liang, Q. Deng, and X. Li, "Teacher–student knowledge distillation for real-time correlation tracking," *Neurocomputing*, vol. 500, pp. 537–546, Aug. 2022.
[55] S. Zhao, T. Xu, X.-J. Wu, and J. Kittler, "Distillation, ensemble and selection for building a better and faster Siamese based tracker," *IEEE Trans. Circuits Syst. Video Technol.*, early access, May 24, 2022, doi: 10.1109/TCSVT.2022.3177215.
[56] T.-Y. Lin, M. Maire, and S. Belongie, "Microsoft COCO: Common objects in context," in *Proc. Int. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2014, pp. 740–755.
[57] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.
[58] E. Real, J. Shlens, and S. Mazzocchi, "YouTube-BoundingBoxes: A large high-precision human-annotated data set for object detection in video," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7464–7473.
[59] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "SiamRPN++: Evolution of Siamese visual tracking with very deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4277–4286.
[60] M. Danelljan, G. Bhat, F. S. Khan, and M. Felsberg, "ATOM: Accurate tracking by overlap maximization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4655–4664.
[61] C. Mayer, M. Danelljan, D. P. Paudel, and L. Van Gool, "Learning target candidate association to keep track of what not to track," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 13424–13434.
[62] C. Mayer et al., "Transforming model prediction for tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8721–8730.



Chen Sun received the B.S. degree from Wuhan University, Wuhan, China, in 2021, where he is currently pursuing the M.S. degree in electronic information with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS).

His research interests include multimodal object tracking and edge computing.



Xinyu Wang (Member, IEEE) received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, in 2019.

Since 2019, he has been an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His major research interests include range over hyperspectral unmixing, detection, restoration, and tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.



Zhenqi Liu received the B.S. and M.S. degrees from the North University of China, Taiyuan, China, in 2012 and 2018, respectively, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

His research interests include computer vision and hyperspectral object tracking.



Yuting Wan (Member, IEEE) received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2017, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

He is currently a Post-Doctoral Researcher with Wuhan University. His major research interests include remote sensing image processing, evolutionary computation, multiobjective optimization, and neural architecture search.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He is currently a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He was a Principal Scientist at the China State Key Basic Research Project, from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He has published more than 700 research papers and five books. He is the Institute for Scientific Information (ISI) highly cited author. He is the holder of 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE Geoscience and Remote Sensing Society (GRSS) 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) student paper contest in recent years. He also serves as an Associate Editor or an Editor for more than ten international journals. He is currently serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is the Founding Chair of IEEE GRSS Wuhan Chapter.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He Organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has published more than 100 research papers in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest and the Single-view Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:25:39 UTC from IEEE Xplore. Restrictions apply.